Department of Information Technology

Academic Year 2025–2026 (Odd Semester)

Degree, Semester & Branch : III Semester, B. Tech, IT
Course Code / Title : CD3291/Data Structures and Algorithms
Name of the Faculty member : Mr. G. Mareeswari

Innovative Practice Description

- **Unit / Topic:** Unit I - Abstract Data Types / Recursion
- **Course Outcome:** CO1
- **Topic Learning Outcome:** TLO4
- **Activity Chosen:** Learning by Teaching
- **Justification:**

Recursion was chosen as the topic for the learning-by-teaching activity because it is a fundamental yet challenging concept in programming that requires deep understanding to master. Students often find it difficult to visualize how recursive calls work, the role of base and recursive cases, and how the call stack manages function executions. By preparing to teach this topic to their peers, students are informed to thoroughly analyze the concept, break it into simpler explanations, and use examples or trace diagrams to illustrate the recursive flow. This process not only strengthens their conceptual clarity but also enhances their problem-solving abilities by applying recursion to examples like factorial computation, Fibonacci series, and sum of n natural numbers.

- **Time Allotted for the Activity:** 35 Minutes
- **Details of the Implementation:**
  - In the previous day, the instructor informed all the students about the activity and insisted to prepare the Recursion concept in advance.
  - The students already learned the concept in the previous semester.
  - On the day of the activity, the instructor randomly selected a student to teach the class, ensuring every student was equally prepared.
  - The selected student explained the fundamentals of recursion, demonstrated examples like factorial computation and Fibonacci series, and used trace diagrams for clarity.

- A question-and-answer session followed the presentation, where peers asked queries and discussed additional insights to reinforce understanding.
- Finally, the instructor summarized the session, highlighted key concepts, and reinforced the applications of recursion in programming and problem-solving.

- **CO–PO/ PSO mapping:**

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO1	2	2	2	1	1	1

(1– Low 2 –Moderate 3 – High)

- **PO/ PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO3**	**PO9**	**PO10**	**PSO1**
	2	**2**	**2**	**1**	**1**	**1**
Justification for correlation	Apply basic Knowledge and fundamentals to understand the recursion concept	Using the concept recursion we can able to analyze and simplify complex Engineering problems.	Able to design and develop solutions for various complex problem.	Students involved in this activity as an individual and also as a team.	Communicate/ share the ideas with other students	Ability to find solution for data science problem using the recursion concept.

- **Images/ Screenshot of the practice:**

Figure 1

Figure 2

Figure 1 and 2 shows the Glimpses of presentation done by Dhanya Sankari. M and Shiri Shabari Sudhan P during the Learning by Teaching Activity.

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    - Students felt confident and motivated when they were given the responsibility to teach their peers.
    - They experienced an increase in communication and presentation skills.
  - ***Benefit of the practice:***
    - Enhances self-learning and critical thinking skills in students.
    - Improves subject retention as teaching reinforces understanding.
    - Promotes peer-to-peer learning and collaboration.
  - ***Challenges faced in implementation:***
    - Some students lacked confidence to teach initially, requiring motivation and guidance.
    - The quality of content delivery varied depending on the student's preparation.
    - Time management in class was a challenge when multiple students presented.

References:

- https://en.wikipedia.org/wiki/Learning_by_teaching
- https://effectiviology.com/protege-effect-learn-by-teaching/
- https://www.researchgate.net/publication/351905566_Impact_of_Seminars_on_Student_Soft_Skills_Development
- https://omerad.msu.edu/teaching/teaching-skills-strategies/27-teaching/162-what-why-and-how-to-implement-a-flipped-classroom-model

Signature of Faculty **Signature of HOD**